GitHub Authentication Guide

PHB 243b: Setting Up Secure Access

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GitHub requires **two-factor authentication (2FA)** for all users. After enabling 2FA, you cannot use your password for command-line Git operations. You must use either **SSH keys** (recommended) or **Personal Access Tokens**.

Step 1: Enable Two-Factor Authentication

1. Go to **Settings** → **Password and authentication** → **Two-factor authentication**
2. Choose your 2FA method:
   - **Authenticator app** (recommended): Google Authenticator, Authy, 1Password, etc.
   - **Security key**: YubiKey or other FIDO2 device
   - **GitHub Mobile**: Use the GitHub app for push notifications
   - **Passkey**: Passwordless authentication (newest option)
3. **Save your recovery codes** in a secure location—you need these if you lose your device

Step 2: Choose Your Command-Line Authentication Method

Option A: SSH Keys (Recommended)

SSH keys provide passwordless authentication and work seamlessly with Git.

Generate a key (run in Terminal):

```
ssh-keygen -t ed25519 -C "your_email@ucsd.edu"
```

Press Enter to accept the default file location. Optionally set a passphrase.

Start the SSH agent and add your key:

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

Copy your public key:

```
cat ~/.ssh/id_ed25519.pub
```

Add to GitHub: Go to **Settings** → **SSH and GPG keys** → **New SSH key**, paste your key.

Test the connection:

```
ssh -T git@github.com
```

You should see: "Hi username! You've successfully authenticated…"

Use SSH URLs when cloning: `git clone git@github.com:username/repo.git`

Option B: Personal Access Token (PAT)

Use this if SSH setup is problematic or you prefer HTTPS URLs.

1. Go to **Settings** → **Developer settings** → **Personal access tokens** → **Tokens (classic)**
2. Click **Generate new token (classic)**
3. Set expiration and select scopes: `repo` (full repository access)
4. Click **Generate token** and **copy it immediately**—you cannot see it again
5. When Git prompts for password, paste your token instead

Store your token using Git Credential Manager to avoid re-entering:

```
git config --global credential.helper store
```

Quick Reference: Which URL Format?

Method	Clone URL Format
SSH	`git clone git@github.com:username/repo.git`
HTTPS + PAT	`git clone https://github.com/username/repo.git`

Check your remote: `git remote -v`

Switch existing repo to SSH:

```
git remote set-url origin git@github.com:username/repo.git
```

Troubleshooting

- **"Permission denied (publickey)"**: Your SSH key is not added to GitHub or ssh-agent
- **"Authentication failed"**: Your PAT expired or has wrong scopes; generate a new one
- **Lost 2FA device**: Use recovery codes at github.com/sessions/recovery

Resources

- GitHub 2FA setup: https://docs.github.com/en/authentication/securing-your-account-with-two-factor-authentication-2fa
- SSH key guide: https://docs.github.com/en/authentication/connecting-to-github-with-ssh
- Happy Git (R users): https://happygitwithr.com/https-pat.html